Zhuoning GUO

Information Retrieval · Multimodal Large Language Models · Agents

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Educations

The Hong Kong University of Science and Technology (Guangzhou) — *Guangzhou, Guangdong, China*

Ph.D. in Artificial Intelligence, Information Hub — *Sep. 2022 - Oct. 2026 (Expected)*

- Supervisor: Prof. Xiaowen Chu (IEEE Fellow), Prof. Renjing Xu

Harbin Institute of Technology — *Harbin, Heilongjiang, China*

B.E. in Software Engineering — *Aug. 2018 - Jun. 2022*

- GPA: 3.6/4.0 (Overall rank: Top 7%)

Internships

AllSpark, Xiaohongshu — *Beijing, China*

Algorithm Research & Development Intern — *May 2026 -*

- **Topics:** Generative search and model behavior alignment
- **Contributions:** Contribute to the production Xiaohongshu Dots agent across post-training, mid-training, and a generative search component:
  - **Post-training:** Contribute to data pipeline development for search question answering and shopping decisions; synthesize tool-augmented, multi-turn agent trajectories that capture tool-use decisions, multi-turn reasoning, and multimodal evidence aggregation, optimizing agent capabilities on complex queries through SFT and RL.
  - **Mid-training:** Explore multi-objective alignment across safety, persona, and product policies; construct model specifications, persona storylines, and rationale-annotated QA data, using semi-supervised learning to internalize multiple capabilities into model parameters.
  - **Generative component:** Lead RedFlip, organizing multimodal search results in pixel-level interfaces and enabling open-ended visual exploration through iterative clicks; integrate it as a callable tool into the Dots agent.

Alibaba Tongyi Lab — *Hangzhou, Zhejiang, China*

Research Intern — *Mar. 2025 - Jan. 2026*

- **Topics:** Multimodal video representation learning
- **Contributions:** Built video retrieval capabilities for multimodal foundation models, spanning data construction, model training, capability evaluation, and research outputs:
  - **Model capability:** Collected, processed, and synthesized 13M+ public samples and completed model training and evaluation; developed a state-of-the-art multimodal video embedding model that improved general-purpose video retrieval by 6.5+%.
  - **Open-source impact:** The work underpinned the video retrieval capabilities of the Qwen3-VL-Embedding series, which surpassed 10M downloads on Hugging Face; released research outputs including papers, datasets, and code.

Baidu Recommendation Architecture Department — *Remote*

Research Intern — *Feb. 2024 - Feb. 2025*

- **Topics:** Information retrieval for web-scale recommender systems
- **Contributions:** Proposed a training–inference co-designed retrieval framework for industrial deployment, improving ANNS performance by 20+%, time on page (ToP) by 1.0+%, and daily active users (DAU) by 0.5+%; deployed across 20+ applications, serving recommendations over 100M+ videos at 100M+ requests per second

Baidu Research — *Haidian, Beijing, China*

Research Intern — *Oct. 2021 - Aug. 2022*

- **Topics:** Spatiotemporal modeling for business intelligence
- **Contributions:** Built a nationwide labor-market forecasting and analytics framework, delivering predictive models, research findings, and proprietary reports

Tencent AI Lab — *Remote*

Research Intern — *May 2021 - Oct. 2021*

- **Topics:** Graph learning for social science
- **Contributions:** Modeled human mobility and contact risk with graph learning to support research on non-pharmaceutical interventions for COVID-19

Microsoft Research Asia (MSRA) — *Haidian, Beijing, China*

Research Intern — *Nov. 2020 - Apr. 2021*

- **Topics:** Computer vision and graphics for design intelligence
- **Contributions:** Developed an intelligent slide-refinement method using a layout autoencoder and aesthetic evaluator, and implemented both the front end and back end of a Microsoft PowerPoint add-in

Publications

[1] **Towards Universal Video Retrieval: Generalizing Video Embedding via Synthesized Multimodal Pyramid Curriculum** *[Paper] [Page]*
Zhuoning Guo, Mingxin Li, Yanzhao Zhang, Dingkun Long, Pengjun Xie, Xiaowen Chu
The 34th ACM International Conference on Multimedia (ACM MM), 2026

[2] **Demystifying Spatial-temporal Scaling Laws of Multimodal Video Embeddings**
Zhuoning Guo, Tingyu Song, Mingxin Li, Yanzhao Zhang, Dingkun Long, Pengjun Xie, Xiaowen Chu
arXiv Preprint, 2025

[3] **GPU-accelerated Multi-relational Parallel Graph Retrieval for Web-scale Recommendations** *[Paper]*
Zhuoning Guo, Guangxing Chen, Qian Gao, Xiaochao Liao, Jianjia Zheng, Lu Shen, Hao Liu
The 35th ACM Web Conference (WWW), 2026

[4] **HiFGL: A Hierarchical Framework for Cross-silo Cross-device Federated Graph Learning** *[Paper]*
Zhuoning Guo, Duanyi Yao, Qiang Yang, Hao Liu
The 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024 **(Oral)**

[5] **Talent Demand-Supply Joint Prediction with Dynamic Heterogeneous Graph Enhanced Meta-Learning** *[Paper]*
Zhuoning Guo, Hao Liu, Le Zhang, Qi Zhang, Hengshu Zhu, Hui Xiong
The 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2022 **(Oral)**

[6] **Labor Migration Modeling through Large-scale Job Query Data** *[Paper]*
Zhuoning Guo, Le Zhang, Hengshu Zhu, Weijia Zhang, Hui Xiong, Hao Liu
The 29th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2025 **(Oral)**

[7] **Against Multifaceted Graph Heterogeneity via Asymmetric Federated Prompt Learning** *[Paper]*
Zhuoning Guo, Ruiqian Han, Hao Liu
The 37th European Conference on Machine Learning and the 30th Conference on Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), 2026

[8] **Discrepant Homophily Co-Preserved Graph Convolutional Network for Labor Migration Forecasting** *[Paper]*
Zhuoning Guo, Le Zhang, Hengshu Zhu, Weijia Zhang, Hui Xiong, Hao Liu
Workshop on Talent Management Computing (TMC) at the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2024

[9] **Convergence-aware Clustered Federated Graph Learning Framework for Collaborative Inter-company Labor Market Forecasting** *[Paper]*
Zhuoning Guo, Hao Liu, Le Zhang, Qi Zhang, Hengshu Zhu, Hui Xiong
arXiv Preprint, 2024

[10] **Rethinking Composed Image Retrieval Evaluation: A Fine-Grained Benchmark from Image Editing** *[Paper]*
Tingyu Song, Yanzhao Zhang, Mingxin Li, Zhuoning Guo, Dingkun Long, Pengjun Xie, Siyue Zhang, Yilun Zhao, Shu Wu
The 64th Annual Meeting of the Association for Computational Linguistics (ACL), 2026

[11] **VimRAG: Navigating Massive Visual Context in Retrieval-Augmented Generation via Multimodal Memory Graph** *[Paper] [Code]*
Qiuchen Wang, Shihang Wang, Yu Zeng, Qiang Zhang, Fanrui Zhang, Zhuoning Guo, Bosi Zhang, Wenxuan Huang, Lin Chen, Zehui Chen, Pengjun Xie, Ruixue Ding
The 43rd International Conference on Machine Learning (ICML), 2026

[12] **Human Mobility Modeling During the COVID-19 Pandemic via Deep Graph Diffusion Infomax** *[Paper]*
Yang Liu, Yu Rong, Zhuoning Guo, Nuo Chen, Tingyang Xu, Fugee Tsung, Jia Li
The 37th AAAI Conference on Artificial Intelligence (AAAI), 2023

[13] **A Cross-View Hierarchical Graph Learning Hypernetwork for Skill Demand-Supply Joint Prediction** *[Paper]*
Wen Shuo Chao, Zhaopeng Qiu, Likang Wu, Zhuoning Guo, Zhi Zheng, Hengshu Zhu, Hao Liu
The 38th AAAI Conference on Artificial Intelligence (AAAI), 2024 **(Oral)**

[14] **Aesthetics++: Refining Graphic Designs by Exploring Design Principles and Human Preference** *[Paper]*
Wenyuan Kong, Zhaoyun Jiang, Shizhao Sun, Zhuoning Guo, Weiwei Cui, Ting Liu, Jian-Guang Lou, Dongmei Zhang
IEEE Transactions on Visualization and Computer Graphics (TVCG), 2022

Visits

The Hong Kong University of Science and Technology *Hong Kong, China*

Visiting Ph.D. Student *Jan. 2024 - Aug. 2024*

- Affiliated to the Department of Computer Science and Engineering during the visiting period

University of Cambridge *Cambridge, England, UK*

Visiting Undergraduate *Jan. 2020 - Feb. 2020*

- Supported by the scholarship from Harbin Institute of Technology (HIT) for the undergraduate visiting program

Awards

2014.11	**First Award, Junior High School, National Olympiad in Informatics in Provinces (NOIP)**, Independent Recruitment Admission Qualification of Fuzhou No.1 High School	*Fujian, China*
2016.11	**First Award, Senior High School, National Olympiad in Informatics in Provinces (NOIP)**, Independent Recruitment Admission Qualification of Harbin Institute of Technology	*Fujian, China*
2020.03	**Finalist (Top 1%), Mathematical Contest in Modeling (MCM)**, Team Leader	*International*
2020.08	**Special Award (Top 4), National IoT Competition (Northeast Region)**, Team Leader	*Northeast China*
2019.10	**First Award, Contemporary Undergraduate Mathematical Contest in Modeling**, Team Leader	*China*

Honors

2025	**Outstanding Reviewer**, SIGKDD 2025	*International*
2019	**Outstanding CLP Member**, Harbin Institute of Technology	*University*
2020	**Outstanding Student Cadre**, Harbin Institute of Technology	*University*

Services

2025	**Session Chair**, SIGKDD 2025	*International*
2022-2025	**Conference Reviewer**, SIGKDD, NeurIPS, ICLR, ICML, WebConf, IJCAI, ICDM	*International*
2024-2025	**Journal Reviewer**, TOIS, TCSVT, Information Science	*International*
2023	**Teaching Assistant**, AIAA 5028: Machine Learning on Graphs at HKUST(GZ)	*University*
2026	**Teaching Assistant**, UFUG 2602: Data Structure and Algorithm Design at HKUST(GZ)	*University*
2020-2022	**Chairman**, HIT Massive Data Club	*Laboratory*
2020-2022	**Captain**, The Football Team of the Faculty of Computing	*Department*
2019-2022	**President**, HIT Chess Club	*Organization*

Volunteers

The Innovation & Entrepreneurship Education Alliance of China *Harbin, Heilongjiang, China*

Conference Volunteer *Oct. 2019*

- Supported the annual national conference hosted by Harbin Institute of Technology, engaging with academic and industry stakeholders.

Rural Primary School *Shangri-La, Yunnan, China*

Teaching Volunteer *Jan. 2019 - Feb. 2019*

- Delivered foundational lessons to students in an underserved rural community, emphasizing clarity and accessibility in knowledge transfer.

Skills

Languages (English)	IELTS 6.5 (5.5) at Aug. 2021
Programming Languages	Python (Recently), C/C++, Java, Bash, C#
Frameworks & Tools	PyTorch (-Lightning), Transformers, Deepspeed, vLLM, Pandas, DGL, PyG, NetworkX, …